

Curation / Classification Checkpoint - 2026-03-07

Purpose

This checkpoint separates the classification work from the UI work.

The classification stream is about:

- trustworthy discrimination between `style_product_decor`, `construction_concern`, `home_maintenance_diy`, and `irrelevant`
- trustworthy multi-axis categorization after the track gate
- preserving prior work as evidence while replacing weak legacy decision layers

The UX stream is important, but it should be handled as a separate cleanup sprint.

Related UX backlog document:

- `docs/UX_CLEANUP_SPRINT_BACKLOG_2026-03-07.md`

Current best runs

These are the current reference runs for v2 classification:

- `track_gate`
 - current run id: `9a308297-d1f1-4509-b29c-d071e2f2d66d`
 - current notes: `raw irrelevant intent override, color-palette beauty carveout, and Leslie magazine clips style signal`
- `multi_axis_inference`
 - current run id: `98f95cbb-70bf-4223-9736-d1a23ecf94dc`
 - current notes: `use source-link evidence for construction concern-domain inference after landscape and inspection review`

What is complete

1. Re-evaluation architecture is in place

The project now has a side-by-side v2 classification layer instead of overwriting legacy outputs.

Implemented already:

- `classification_runs`
- `asset_field_provenance`
- `asset_track_assessments`
- `asset_axis_memberships`
- `asset_axis_evidence`
- `asset_overrides`

This is consistent with the migration design in:

- `docs/CURATION_REEVALUATION_MIGRATION_SPEC_2026-03-06.md`

2. Title provenance and working-title workflow are in place

This is important because title quality was part of the earlier compounding-error problem.

Current title handling now distinguishes between:

- the current working title
- a better original/source-derived title where possible
- a concise suggested title for workflow use

3. v2 track gate is live and usable

Latest counts:

- `style_product_decor` = 4301
- `construction_concern` = 189
- `home_maintenance_diy` = 7
- `irrelevant` = 172
- `ambiguous` = 282

This is a major improvement over the older curation logic, especially around the previous construction over-bucketing problem.

4. v2 multi-axis inference is live and usable

The classifier now writes non-exclusive memberships such as:

- `space_context`
- `subject_type`
- `room`
- `product_focus`
- `concern_domain`
- `product_system_focus`

This is the key structural change that prevents forcing everything into a room bucket.

5. Review exports now exist for the remaining problem slices

A repeatable operational export has been added:

- `tools/export_classification_review.py`

Latest export directory:

- `data/exports/classification_review_checkpoint_20260307/`

Generated files:

- manifest.json
- ambiguous_track.json
- ambiguous_track.csv
- pinterest_construction.json
- pinterest_construction.csv
- maintenance_diy_track.json
- maintenance_diy_track.csv
- pinterest_maintenance_diy.json
- pinterest_maintenance_diy.csv
- undifferentiated_envelope.json
- undifferentiated_envelope.csv

Current metrics

Track counts

- current:
 - style_product_decor = 4301
 - construction_concern = 189
 - home_maintenance_diy = 7
 - irrelevant = 172
 - ambiguous = 282

Remaining review slices

- current:
 - ambiguous_track = 282
 - ambiguous_low_signal_url = 20
 - ambiguous_media_mismatch = 135
 - ambiguous_media_link_mismatch = 30
 - ambiguous_media_weak_thumbnail = 96
 - ambiguous_true_contested = 127
 - pinterest_construction = 24
 - maintenance_diy_track = 7
 - pinterest_maintenance_diy = 3
 - undifferentiated_envelope = 0
 - source_link_conflicting = 158
 - source_link_insufficient = 967

Why these slices matter

These are the highest-value slices for tightening the classifier further:

1. ambiguous_track
 - these are the items the gate itself is least sure about
2. pinterest_construction

- this is now the narrowed true-construction residual for the Pinterest review set
3. `pinterest_maintenance_diy`
 - this is the new adjacent-bucket slice created from the reviewed Pinterest maintenance/repair items
 4. `maintenance_diy_track`
 - this is the broader cross-source adjacent bucket that still needs spot-checking for precision
 5. `undifferentiated_envelope`
 - this is the remaining catch-all bucket we want to reduce further

What the exports show

Step 2 tightening pass outcome

The latest pass targeted two failure modes:

1. generic DIY / maintenance / utility content that should not stay in `construction_concern`
2. true envelope concerns that were real construction items but still too broad in `concern_domain=envelope`

Implemented changes:

- added stronger irrelevant signals for:
 - cleaning
 - mopping
 - lifehacks
 - moving furniture
 - stripped screw / household-product / woodworking-style utility content
- added a `diy-specific` irrelevant override when the evidence looks like generic non-home DIY rather than a house-planning concern
- expanded construction `product_system_focus` to include more specific envelope systems such as:
 - `door_system`
 - `garage_door_system`
 - `siding_system`
 - `flashing_system`
 - `sheathing_system`
 - `roof_vent_system`
 - broader `window_system` coverage

Result:

- obvious false positives such as:
 - Living Off Grid - A Home Made Deep Well Pump
 - How We Got Our Stained Grout White Again

- Power Tool Woodworking for Everyone Online Overarm Pin Router
- the Facebook shower-cleaning reel now land in **irrelevant**
- true envelope construction items now pick up specific **product_system_focus** memberships instead of staying broad-only

1. Ambiguous-track cases still contain low-signal drift

Representative examples show that a number of ambiguous Facebook items are still landing in **style_product_decor** with very weak evidence, sometimes mostly because of inherited **assets.category=home_design**.

Examples from the export:

- Driveway sweeper Part 01
- Adkins Materials
- Association of Professional Builders (APB): Builders Qualifying Checklist

Interpretation:

- the classifier is no longer wildly overconfident here, which is good
- but the residual default/style prior is still too generous in some low-signal cases

2. Pinterest construction still includes utility/how-to contamination

Representative examples from the export:

- Living Off Grid - A Home Made Deep Well Pump - Mother Earth News
- How We Got Our Stained Grout White Again | Young House Love
- Power Tool Woodworking for Everyone Online Overarm Pin Router

Interpretation:

- some of these are plausibly construction-adjacent
- some are really maintenance, utility, DIY, or workshop content rather than house-planning concerns
- this slice needs better negative evidence and better handling of generic how-to content

3. Undifferentiated envelope still contains obvious noise

Representative examples from the export:

- real construction videos in **building**
- partially-understood under-construction scenes
- an obvious false positive from **Cleaning Tips**

Interpretation:

- the broad old `Envelope = 913` failure mode has been dramatically reduced
- but `envelope` is still too easy to trigger without a named system or stronger building-envelope evidence

This statement is now partially superseded by the latest pass:

- `undifferentiated_envelope` is currently 0

That does not mean the construction axis work is finished. It means the current remaining problem is less about broad envelope collapse and more about validating whether the new `product_system_focus` values are the right ones.

4. Pinterest regrouping pass is now aligned with the reviewed proposal

The Pinterest construction slice was reviewed manually and grouped into four intended outcomes:

1. `style reference`
2. `construction concern`
3. `construction-adjacent maintenance / repair / utility`
4. `irrelevant / noisy misfire`

The latest classifier pass now matches that proposal exactly for the reviewed 29-item subset:

- 13 / 13 proposed `style reference` items now land in `style_product_decor`
- 8 / 8 proposed `construction concern` items remain in `construction_concern`
- 5 / 5 proposed `maintenance / repair / utility` items now land in `home_maintenance_diy`
- 3 / 3 proposed `irrelevant / noisy misfire` items now land in `irrelevant`

Implication:

- the remaining Pinterest construction slice is no longer a mixed plan/reference bucket
- the construction and maintenance-adjacent material are now separated cleanly in this reviewed slice

5. Maintenance / DIY is now a real fourth track

This was a taxonomy decision, not just a scoring tweak.

The classifier now has a first-class adjacent track:

- `home_maintenance_diy`

Current status:

- `total home_maintenance_diy = 17`
- `pinterest_maintenance_diy = 5`
- the reviewed Pinterest maintenance set now lands there exactly as intended

Current caveat:

- the cross-source bucket is much smaller and cleaner than the first draft, but it still needs spot review for precision, especially on Facebook

Classification work that remains

1. Tighten low-signal track decisions

Priority:

- reduce cases where weak or inherited priors allow a low-information item to stay in `style_product_decor`

Likely work:

- reduce the fallback power of `assets.category=home_design`
- require stronger positive evidence before keeping ambiguous non-visual items on the style track
- push clearly generic content more readily toward `irrelevant`

2. Tighten cross-source maintenance / DIY precision

Priority:

- validate that the new adjacent bucket is precise outside the manually-reviewed Pinterest set

Likely work:

- inspect the `maintenance_diy_track` export visually
- remove any remaining generic tool/garden/problem-solving drift
- decide whether this bucket needs its own future axis model or can continue borrowing the construction-side axes

3. Tighten Pinterest construction

Priority:

- separate real construction concerns from generic maintenance / DIY / workshop / utility content

Likely work:

- add negative evidence for generic how-to and utility content
- downweight workshop/tool content unless there is explicit house-system evidence

- distinguish `construction concern` from `home maintenance tip`

Current status after the latest pass:

- improved from 38 to 8
- the false-positive plan/reference cluster has been cleared
- the maintenance/repair subset has been split into its own adjacent track

The remaining rows still appear to contain a mixture of:

- true house-planning concerns
- build-system references
- checklist/planning material
- a smaller amount of ambiguous residue

4. Tighten undifferentiated envelope

Priority:

- avoid using `envelope` as a fallback concern domain when the evidence is weak

Likely work:

- require explicit system or building-envelope evidence
- downweight bathroom/cleaning/shower drift unless there is clear enclosure/waterproofing context
- prefer `unknown/ambiguous` over a broad envelope label when evidence is mixed

Current status after the latest pass:

- the raw `undifferentiated_envelope` slice is currently 0

So the next issue is no longer broad envelope collapse itself.

The next issue is validating the specificity and usefulness of the new envelope system labels.

5. Start consuming v2 axes in report generation

This remains a necessary follow-through step.

Report generation should use:

- `space_context`
- `subject_type`
- `room` only when justified
- `product_focus` / `product_system_focus`

The important rule remains:

- do not force room assignment unless the asset is actually an interior full-room scene or equivalent strong room evidence exists

6. Review landscape items for style-vs-construction split

Priority:

- separate landscape mood/reference material from actual lot-landscaping construction scope

Why this is needed:

- **landscape** is overloaded
- some items are aesthetic precedent and belong in **style_product_decor**
- some items represent real site-work or lot-landscaping tasks and belong in **construction_concern**

Working rule:

- keep items in **style_product_decor** when they are about how the grounds should feel or look
- move items to **construction_concern** when they are about work that must be executed on the lot or site

Likely work:

- review landscape/reference items currently landing on the style side
- spot-check whether any of them are actually site-work / execution concerns
- tighten the wording and examples so future review passes treat **landscape mood** and **lot landscaping task** as different things

Current status after the focused review:

- the dedicated landscaping follow-up collection is complete
- final reviewed split:
 - **construction_concern** = 24
 - **style_product_decor** = 3
- implication:
 - the landscape ambiguity is real but bounded
 - most of the reviewed slice was legitimately lot / site / execution work, not style mood/reference

What is explicitly deferred to the UX sprint

These items should not drive classifier changes directly.

Examples:

- side panel organization and discoverability
- one-by-one review ergonomics
- modal layout and wording
- 3D Explorer control clarity
- visual polish and interaction cleanup

Operational note:

- the recent count-sync bug in one-by-one review has been fixed so the live scope, sidebar node count, and hidden count move together correctly
- that fix is operationally useful, but it is not part of the core classifier checkpoint

Recommended next classification step

The next engineering pass should operate on the new review exports rather than on broad intuition.

Recommended order:

1. inspect the updated `pinterest_construction.csv`
2. inspect the new `pinterest_maintenance_diy.csv` and `maintenance_diy_track.csv`
3. review landscape items that may be style mood/reference versus lot-landscaping construction scope
4. inspect the highest-noise rows in `ambiguous_track.csv`
5. validate the new `product_system_focus` values for envelope-heavy construction items
6. once that is stable, start consuming v2 axes in report generation

Checkpoint artifacts

Primary design docs:

- docs/CURATION_CONFIDENCE_HIERARCHY_SPEC_2026-03-05.md
- docs/CURATION_PROVENANCE_AUDIT_2026-03-05.md
- docs/CURATION_REEVALUATION_MIGRATION_SPEC_2026-03-06.md
- docs/CURATION_V2_PROGRESS_2026-03-06.md

Operational review artifacts:

- data/exports/classification_review_checkpoint_20260307/manifest.json
- data/exports/classification_review_checkpoint_20260307/ambiguous_track.csv
- data/exports/classification_review_checkpoint_20260307/pinterest_construction.csv
- data/exports/classification_review_checkpoint_20260307/pinterest_maintenance_diy.csv
- data/exports/classification_review_checkpoint_20260307/maintenance_diy_track.csv
- data/exports/classification_review_checkpoint_20260307/undifferentiated_envelope.csv

2026-03-09 addendum

Construction review pass completed

Jim completed a manual one-by-one review pass over the Track -> Construction slice.

Recent active overrides persisted during that pass window:

- `construction_concern` = 76
- `style_product_decor` = 10

- irrelevant = 5
- home_maintenance_diy = 4
- total persisted in the pass window: 95

Current active Jim track overrides overall:

- construction_concern = 158
- style_product_decor = 21
- irrelevant = 11
- home_maintenance_diy = 5
- total active Jim track overrides: 195

Interpretation:

- the top-level four-track workflow is usable enough for real manual review
- the next classification pressure point is no longer broad construction triage
- it is the sub-structure around landscaping and inspection-like material

Landscaping follow-up is now the main focused review slice

During the construction review pass, Jim used the reason field to mark landscaping-related items.

Current landscaping-note counts across active Jim track overrides:

- 27 items mention **landscap**
- current track split within that slice:
 - construction_concern = 23
 - style_product_decor = 4

A follow-up collection was created to make that review slice accessible in the app:

- Review: Landscaping Follow-Up - 2026-03-09
- collection id: ba65d193-5dc4-4650-ae1a-be77db2e85e4
- item count: 27

This was intentionally a mixed slice.

It answered:

- which items are landscape mood/reference and should stay on the style side
- which items are site-work / drainage / irrigation / lot-landscaping concerns and should stay on the construction side

Final reviewed result:

- construction_concern = 24
- style_product_decor = 3

Inspection is now a live construction concern domain

`inspection_quality_control` is now a first-class `concern_domain` in the v2 model.

Latest axis count:

- `inspection_quality_control = 2`

What changed:

- the focused inspection review confirmed the need
- the classifier now maps strong inspection evidence into `inspection_quality_control`
- stored source-link text is now used in construction-axis inference, which makes source-backed inspection/build topics usable even when the thumbnail/title are weak

One unresolved wrinkle remains:

- asset `5af0e064-01fd-43c9-91a1-4c4dc4ed5852` is still pinned to:
 - `concern_domain = plans_code_permits`
 - `project_phase = permit_code`
- reason:
 - it carries older active manual axis overrides from the earlier workflow
 - those overrides were preserved rather than automatically removed

One-by-one review is now materially safer

The overnight UX cleanup completed these review-mode changes:

- one-by-one review and modal review now use the same track-review model
- the review card now shows the asset ID
- the one-by-one review form keeps a local draft while Jim types
- undo/go-back restores the saved review draft instead of dropping the typed reason
- **Irrelevant** is now a dedicated action button rather than a duplicated dropdown choice
- both review surfaces now include a structured **Review focus** (optional) field:
 - **Landscaping**
 - **Inspection**

Validation completed:

- `node --check app/app.js`
- `PYTHONPATH=src python3 -m unittest -v tests.test_server_api`
- result: 65 tests passed

Next starting point

Recommended order for the next session:

1. open the app
2. use the review workflow on the grouped source-link conflict slice
3. decide whether to clear the one stale inspection-axis override that still pins an item to **plans_code_permits**
4. once the source-link conflict slice is smaller, resume report-generation work on top of the v2 axes